Hsinpin Wen

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Education

Master of Science in Biostatistics Aug 2026 – May 2028
Johns Hopkins Bloomberg School of Public Health

Bachelor of Science in Statistics and Data Science, Minor in Spatial Studies Sept 2022 – Mar 2026
University of California, Santa Barbara

Experience

Undergraduate Research Assistant, MOVE Lab Oct 2025 – Dec 2025
Santa Barbara, CA

- Developed a tiger trajectory simulation using step-selection functions to model movement behavior and quantify the effects of environmental covariates from GPS telemetry data across spatial and temporal contexts.
- Analyzed large-scale spatial and environmental datasets using R, Python, and ArcGIS Pro to identify movement patterns, behavioral features, and habitat associations relevant to computational ecological modeling.
- Built and evaluated predictive and geostatistical models to assess trajectory dynamics, model performance, and sensitivity to changing environmental conditions.
- Interpreted model outputs and visualizations to communicate statistical findings, methodological insights, and reproducible research results to collaborators and research partners.

Data Science Intern, National Center for Ecological Analysis and Synthesis May 2024 – Sep 2025
Santa Barbara, CA

- Used R to curate metadata and archive ecological, physical, and social datasets from more than 200 NSF-funded research projects for the Arctic Data Center.
- Developed and improved data processing workflows to strengthen quality control, metadata standardization, and operational efficiency across large collaborative research projects.
- Coordinated with scientists nationwide to acquire, clean, and format research data, ensuring data integrity and consistency for downstream statistical analysis and reuse.
- Collaborated with software engineers and programmers to test database tools, evaluate infrastructure updates, and provide feedback to improve system functionality and performance.
- Mentored new interns by explaining workflows, answering technical questions, and supporting data curation and quality assurance tasks.

Data Analyst Intern, OCBridge Jun 2025 – Aug 2025
San Jose, CA

- Developed automated Python pipelines to extract, structure, and manage large-scale online datasets, improving processing efficiency and supporting reproducible analytics workflows.
- Cleaned, validated, and standardized candidate and company data to improve dataset quality, ensure consistency, and support machine learning and statistical modeling applications.
- Implemented quality control checks across heterogeneous datasets to identify inconsistencies, maintain data integrity, and facilitate accurate reporting and analysis.

Undergraduate Research Assistant, Trugman Lab Oct 2024 – Jun 2025
Santa Barbara, CA

- Conducted numerical and statistical modeling of field-collected tree data to assess how environmental factors influence forest health, resilience, and long-term ecosystem dynamics across varied ecological conditions.
- Wrote and implemented Python scripts to clean, transform, and organize complex ecological datasets, ensuring data integrity, reproducibility, and readiness for downstream statistical modeling.
- Evaluated model outputs and data visualizations to identify key environmental predictors of tree health and communicate statistical findings and research insights to collaborators.

Skills

- **Programming:** Python, R, SQL, Base SAS, MATLAB
- **Statistical Methods & Coursework:** Statistical modeling, experimental design, Bayesian data analysis, survival analysis, simulation studies, regression analysis, probability, statistical inference
- **Data Science Tools:** Git/GitHub, Bash, Jupyter Notebook, RMarkdown, Quarto, LaTeX, Microsoft Office